

Ganeti introduction

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What is Ganeti?

- virtualization cluster manager (KVM)
- local distributed storage (DRBD)
- high available, no SPOF, scale out
- python3 / Haskell
- is „selfcontained”
- 16.07. 2007: initial commit

What is (full) virtualization (KVM)?

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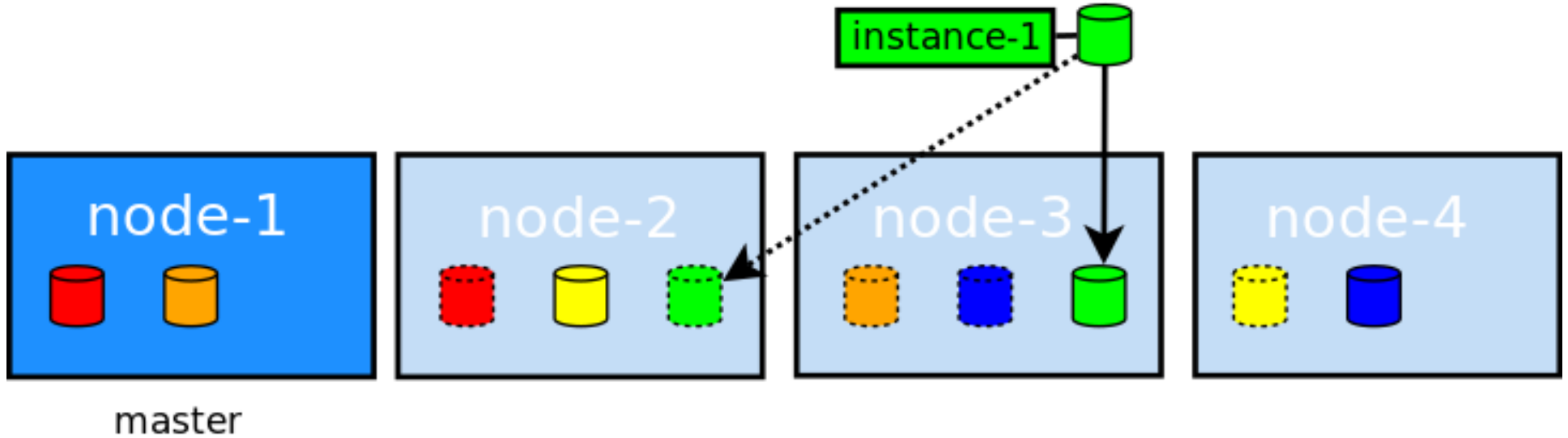
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- not containers (lxc, docker, k8s, ...)

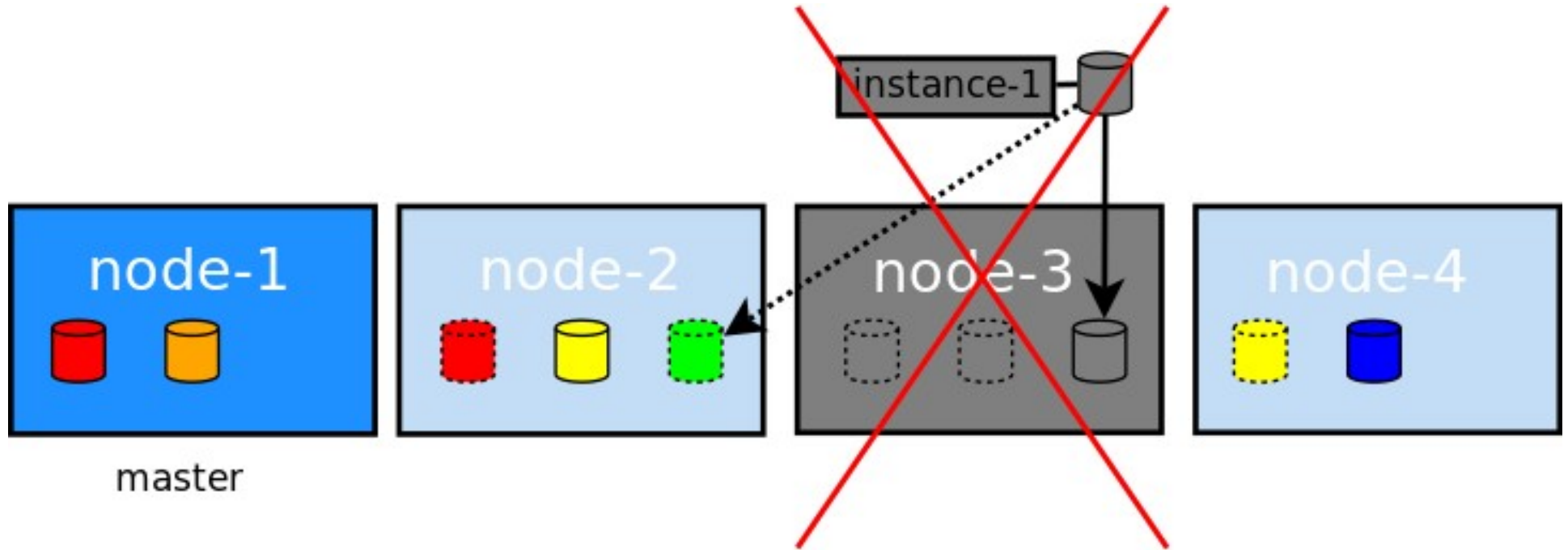
Ganeti DRBD architecture



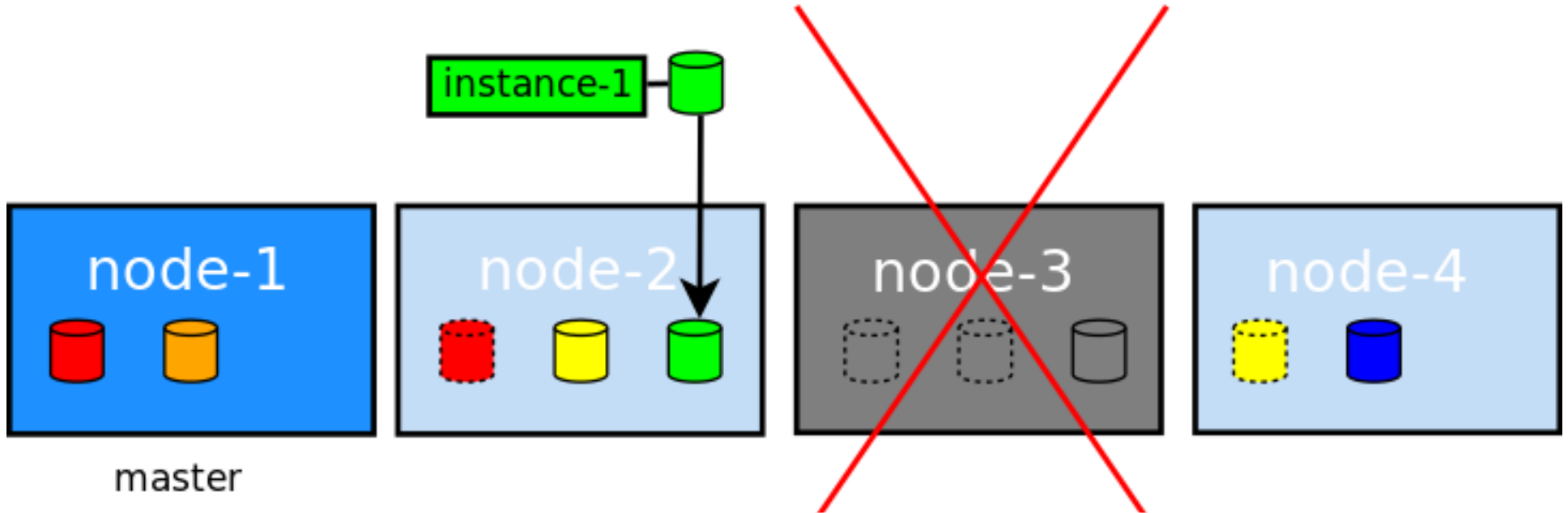
Cluster:

- manage multiple hosts (nodes)
- achieve N+1 node redundancy

Ganeti DRBD architecture

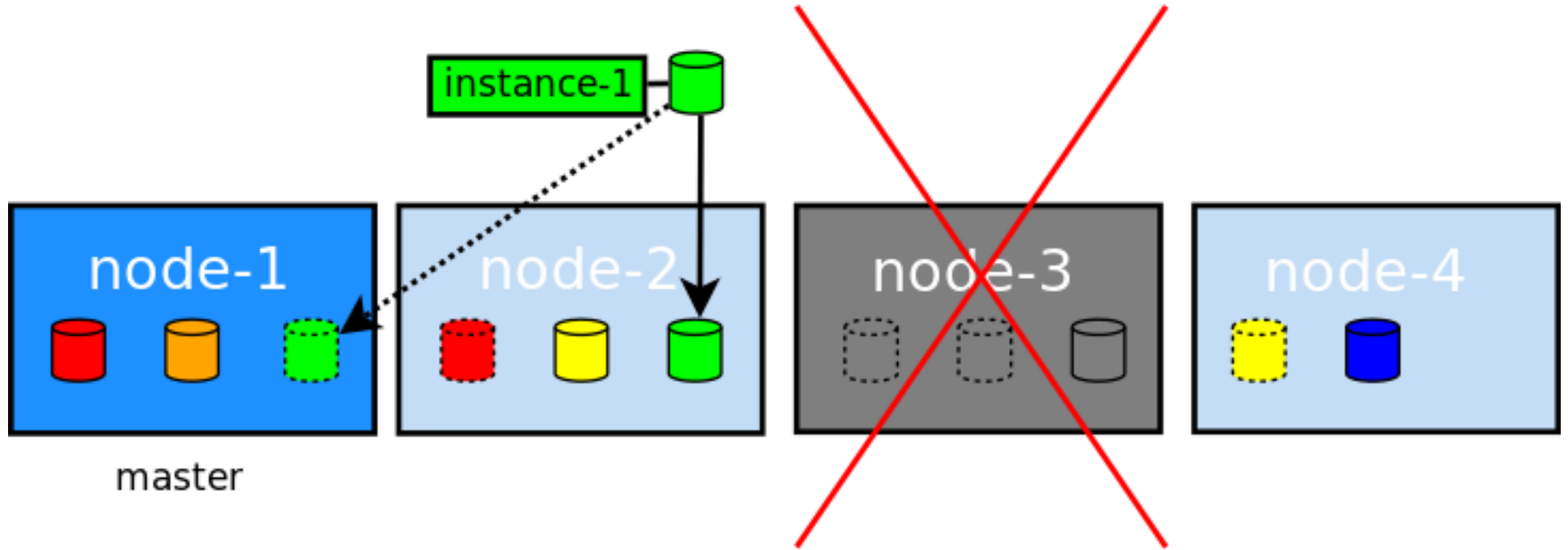


Ganeti DRBD architecture

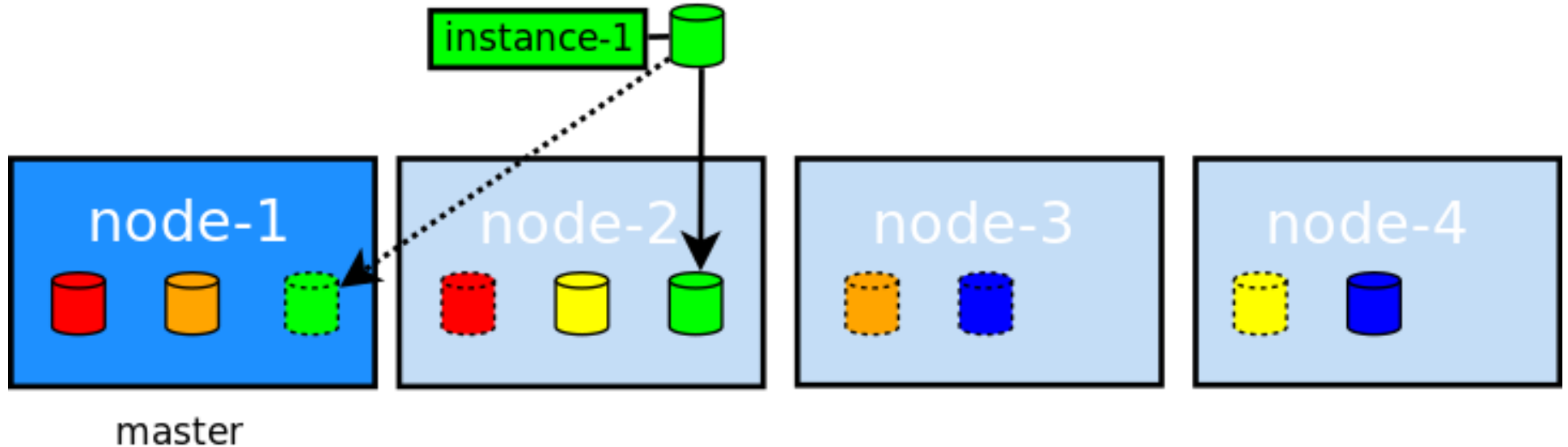


- N+1 redundancy guaranteed (RAM)

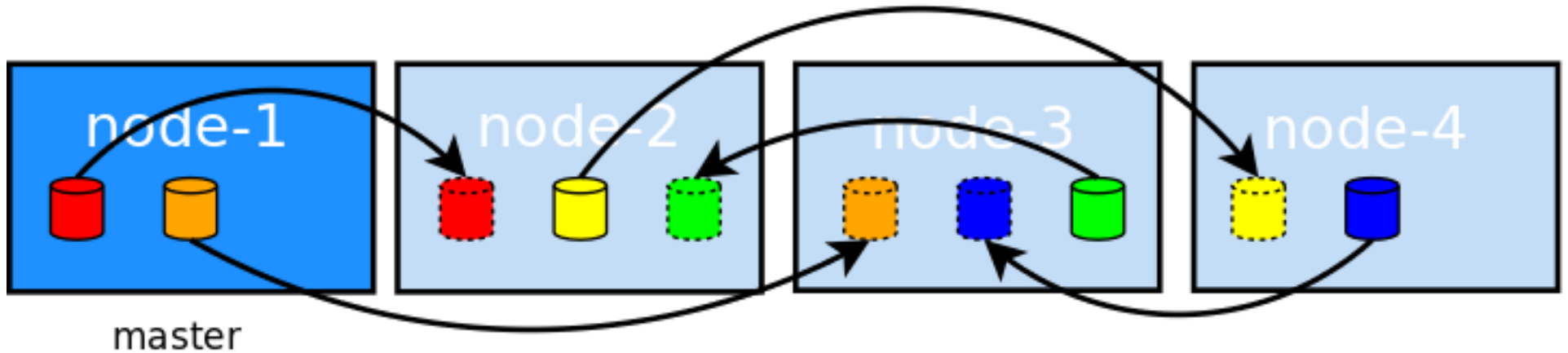
Ganeti DRBD architecture



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Ganeti DRBD connections



- 4 node cluster
- 5 instances: red, orange, yellow, green, blue
- single DRBD con 2 nodes → many VMs mesh

Ganeti htools (Haskell)

- placing instances's pri/sec disks matters
- bin packing, but with dependency on 2nd node
- Iustin Pop @ICFP 2010, Baltimore

<https://k1024.org/papers/icfp10-haskell-reagent.pdf>

- *“placement algorithm is not as straightforward”* (because of dependency on 2nd node)
- *“adoption barrier is quite high, Haskell is a good asset for solving certain types of problems”*

Ganeti htools (Haskell)

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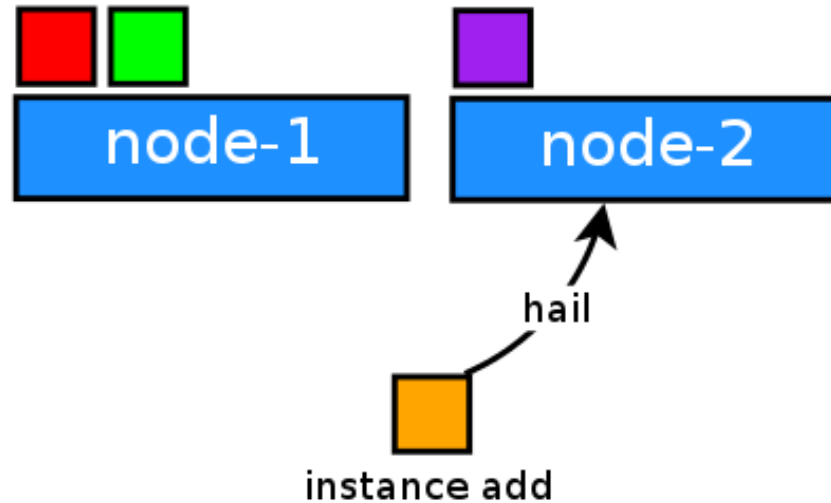
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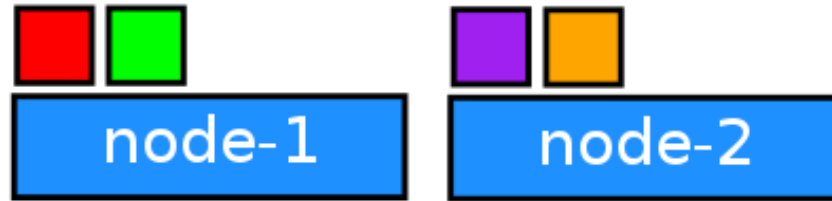
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- manual placement difficult
- at no point manual intervention needed

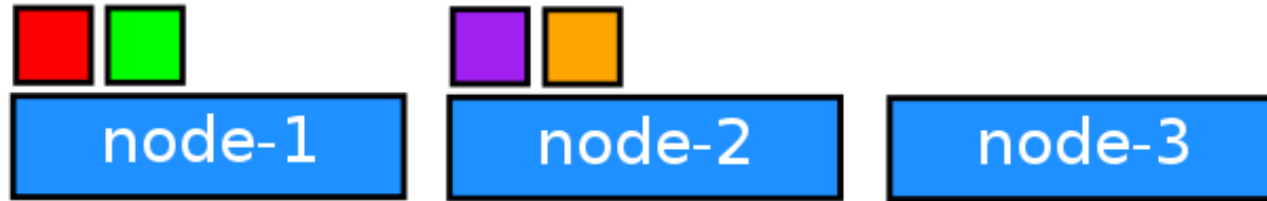
Ganeti htools (Haskell)



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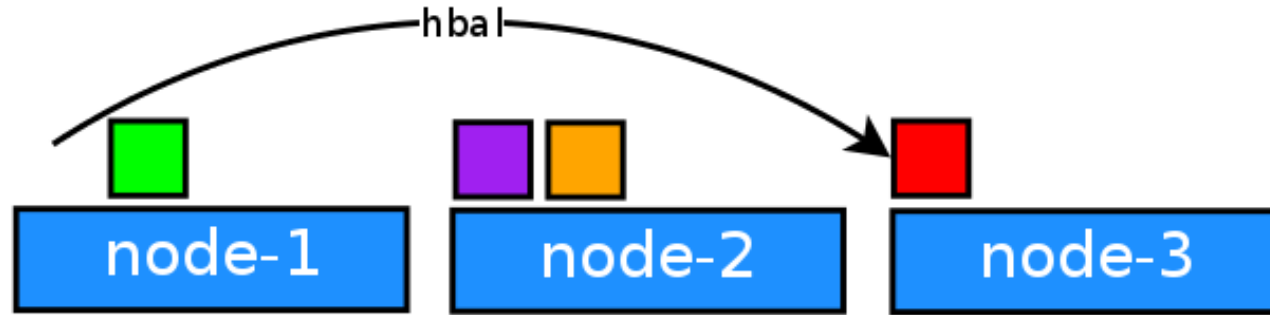


Ganeti htools (Haskell)



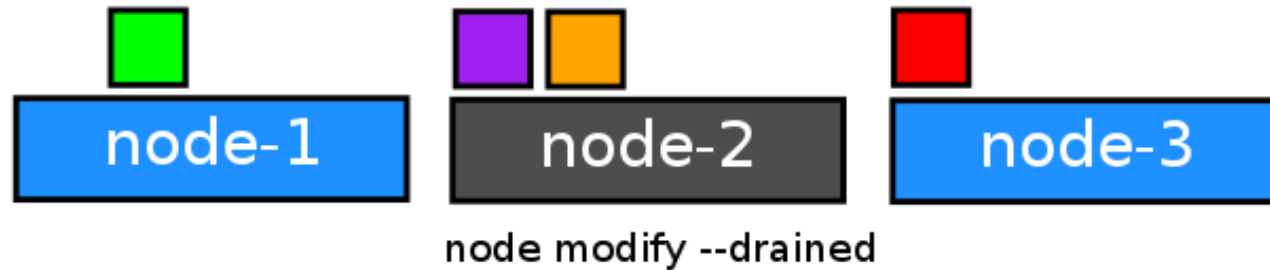
node add

Ganeti htools (Haskell)



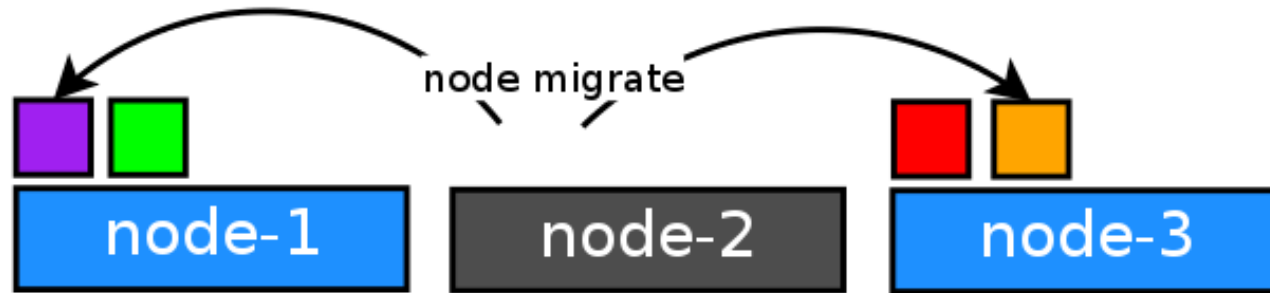
rebalance

Ganeti htools (Haskell)



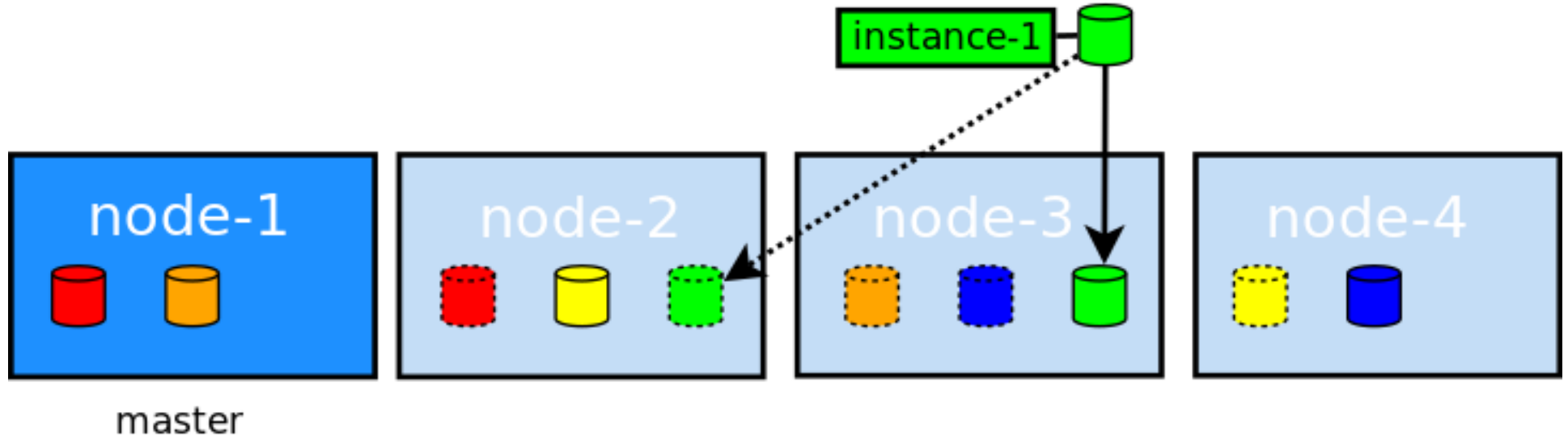
drained=no allocation

Ganeti htools (Haskell)

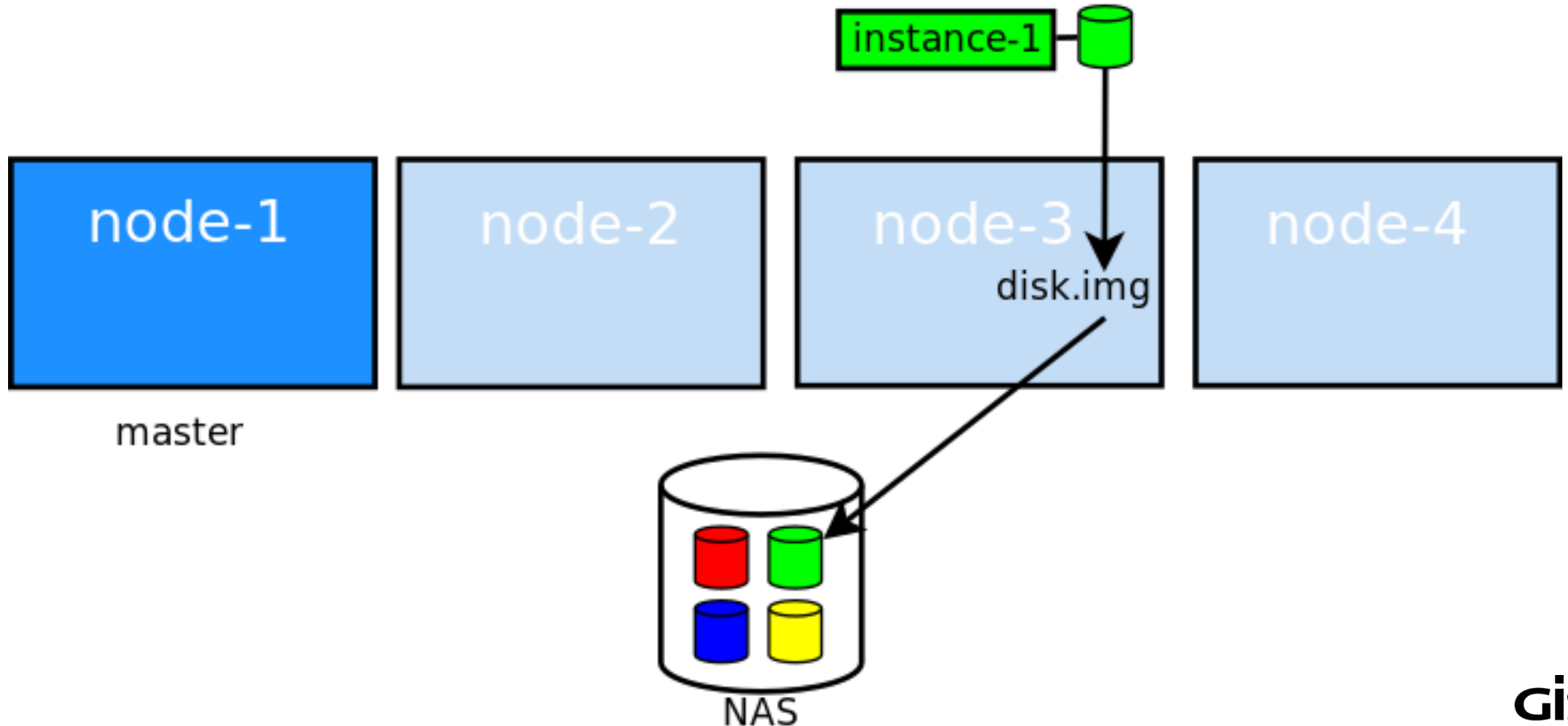


node evacuate

Scale out storage in OSS for HCI



External: no HCI, no scale-out



Scale out storage in OSS for HCI

IBM/Redhat  ceph	Linux  (8.4)

There is also GlusterFS , bought by Redhat 23 month before buying Ceph

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configurable redundancy (typical +2)	+1 (aka RAID1)
all VMs lost if enough nodes fail	only VMs lost that have their primary and secondary disks on lost nodes
VM disks only accessible through ceph client	VM disks are LVM volumes, can be accessed without DRBD

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higher latency	local reads, writes +1 network RTT
no RAID on local disks	RAID on local disks advised

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Ganeti comparison

- cons:
 - does not have a UI (different roles / permissions)
 - lacks features (snapshot, backup, PCI p/t, ...)
 - is widely unknown / has a very small community
- pros:
 - selfcontained (no DB etc. needed)
 - unique handling of DRBD / N+1 resource model
 - community driven and part of Debian

Ganeti comparison

- fits well:
 - if a single person / team manages nodes & instances
 - you like CLI and RestAPI
 - you want HCI, but fear Ceph complexity / latency
 - your nodes run Debian/Ubuntu (also RHEL based)
 - you focus on availability, uninterruptible node maintenance/scaling and rolling node OS upgrades
- Used by: wikimedia, Debian, sipgate, GISA, ...

Other players

- home: virt-manager: GUI, single node
- datacenter virtualisation (also Ganeti)
 - Ovirt: Redhat centric, RHEV community, EOL
 - **Proxmox**: not so new star, GUI, HCI (Ceph)
 - ??? lxd/incus, opennebula, Apache CloudStack
- Cloud
 - Openstack: full blown IaaS/cloud (i.e. OVHcloud)
 - Custom solutions: i.e. Hetzner Cloud

Compare the right layer / match with your needs



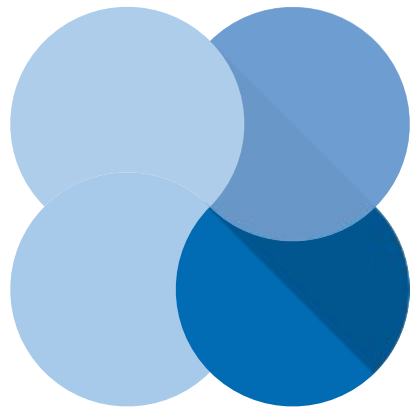
FOSDEM 2013
vkoukis@grnet.gr

OPENSTACK	SYNNEFO	UI	
OPENSTACK	OPENSTACK	API	
OPENSTACK	SYNNEFO	CLOUD	
----- OPENSTACK -----			
		CLUSTER	
----- LIBVIRT -----	----- GANETI -----		
LIBVIRT		NODE	
KVM	KVM	HYPERVISOR	

End

Questions?

<https://ganeti.org>



ganeti

Backup: GISA numbers

Instances view:

number of instances: 651

total vCPUs assigned to instances: 3863

total RAM assigned to instances: 12666 GiB

total diskspace assigned to instances: 212574 GiB

Node view:

number of nodes: 26

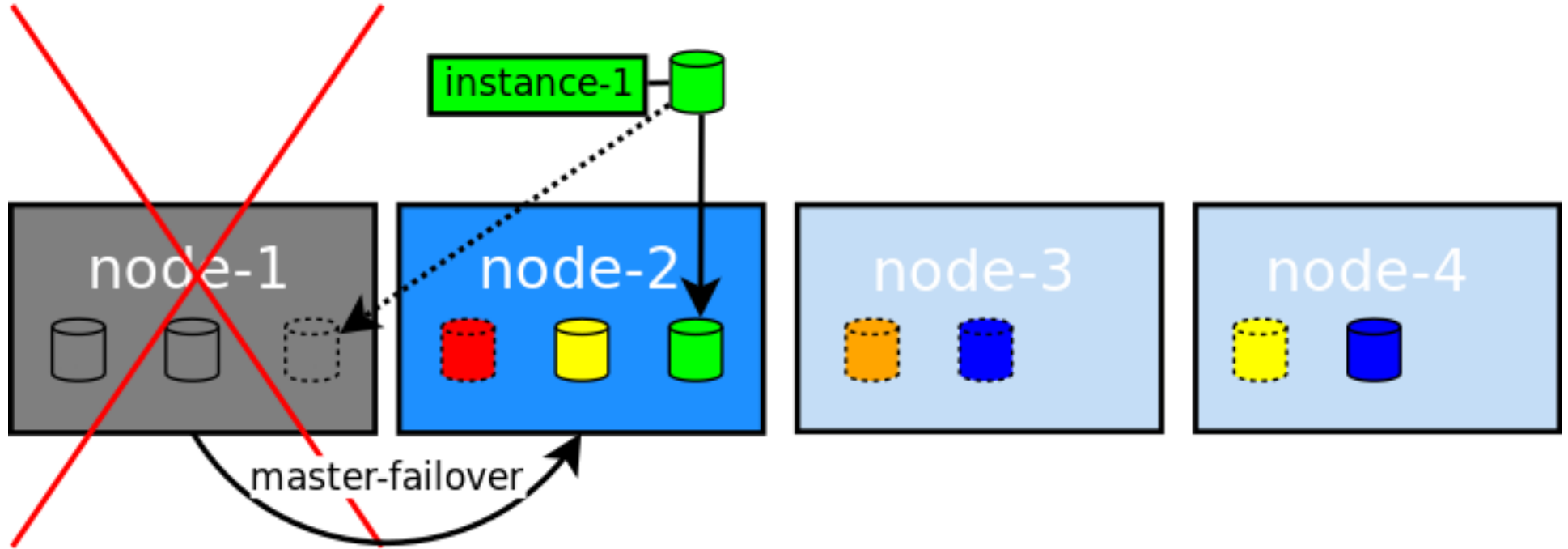
Backup: Htools algorithm

- objective function is cluster score:
 - weighted sum of standard deviation of:
 - percent free memory
 - percent reserved memory
 - percent free disk
 - ratio virtual-to-physical cpus
 - count of nodes failing N+1
 - ...

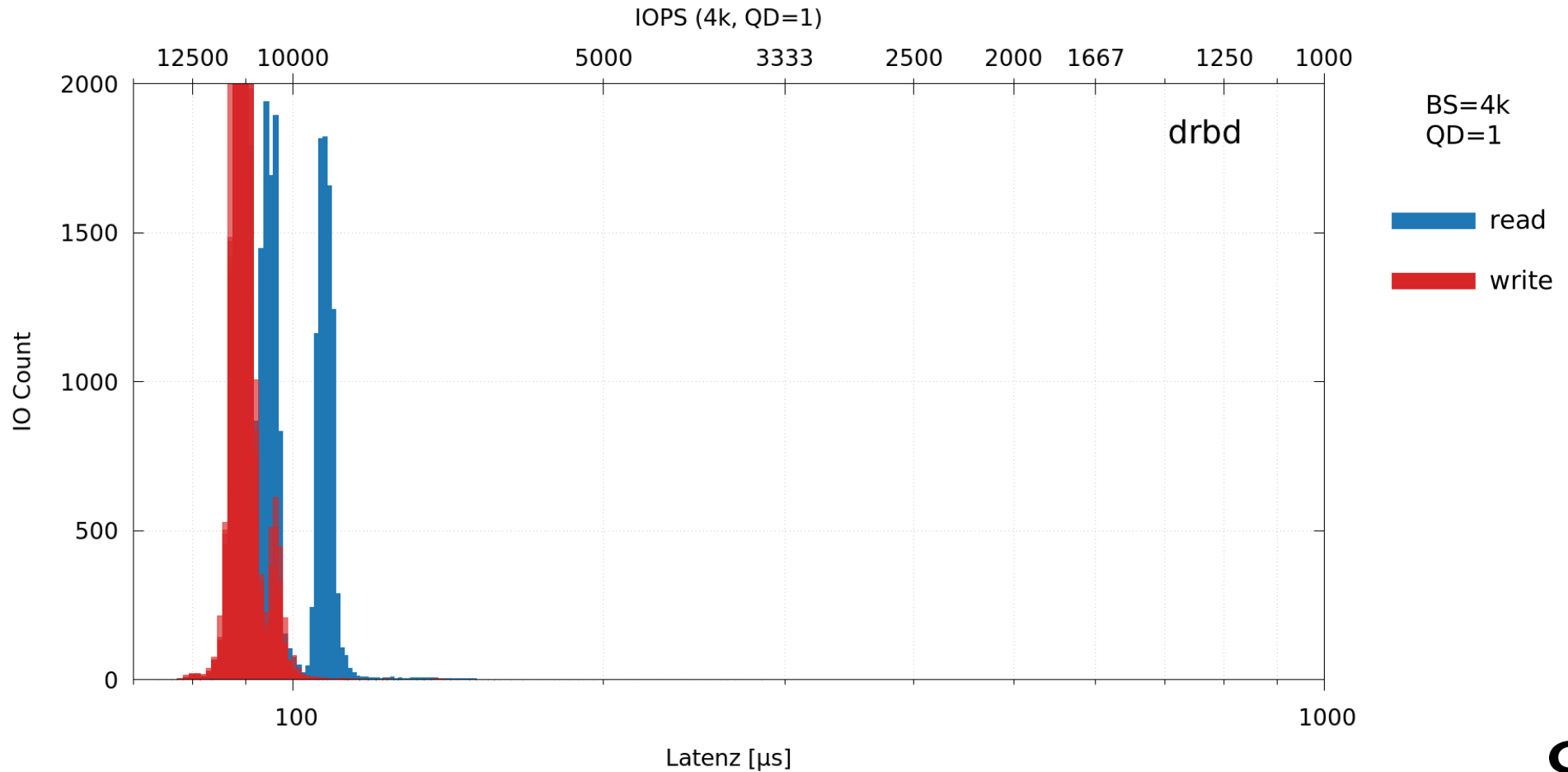
Backup: Htools algorithm

- independent steps
- at each step, compute best instance move that lowers the cluster score (gradient descent)
- moves:
 - failover (f)
 - replace secondary (r)
 - replace primary, a composite move (f, r, f)
 - failover and replace secondary, also composite (f, r)
 - replace secondary and failover, also composite (r, f)

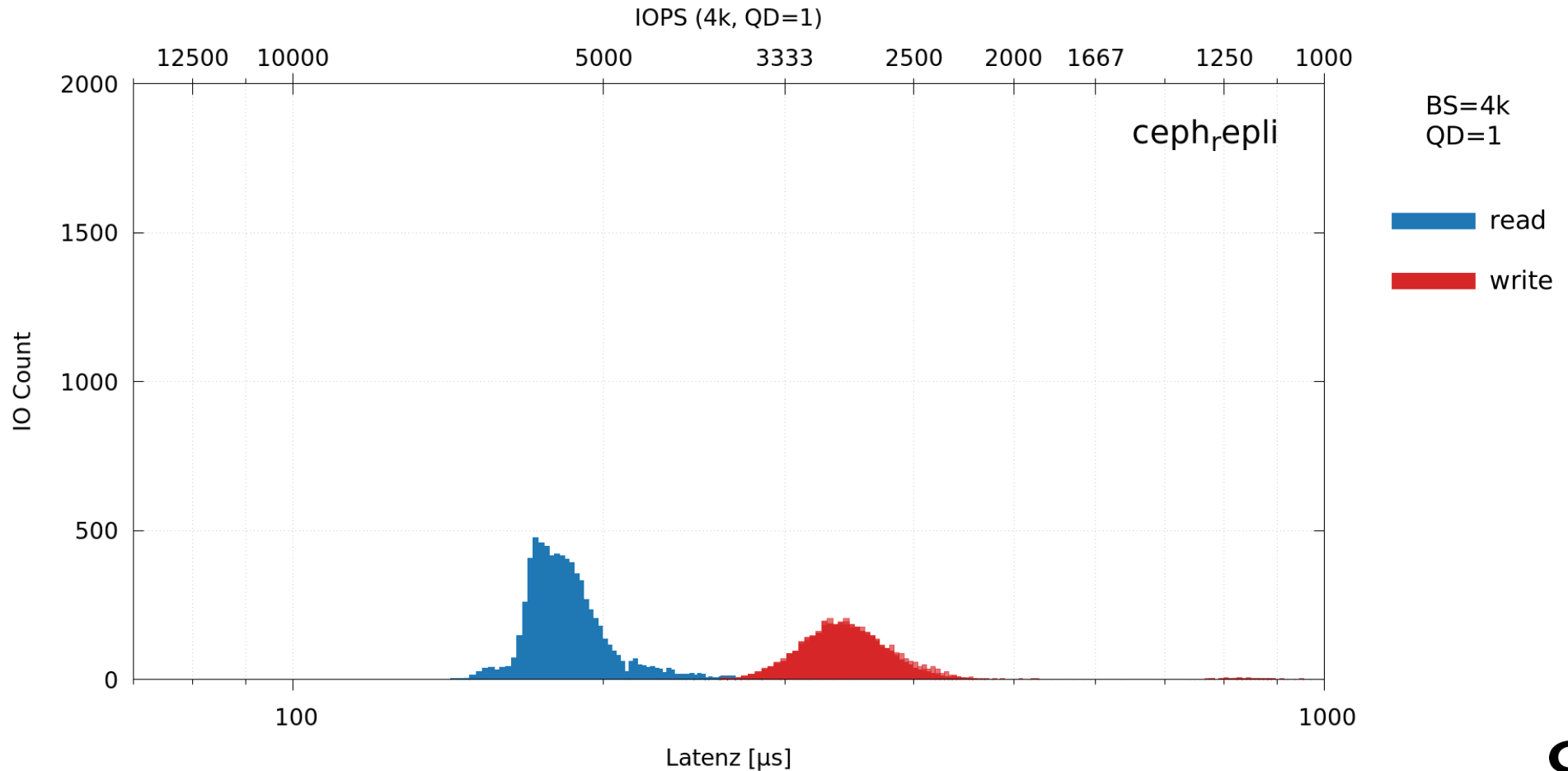
Backup: Ganeti DRBD architecture



Backup: DRBD latency



Backup: Ceph 3x replication latency



Backup: Ceph EC4+2 latency

